



Killara
HIGH SCHOOL

MATHEMATICS DEPARTMENT

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Centre Number

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Student Number

2019

Year 12 Extension 1 Mathematics

Task 3 18th June 2019

General

Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:

44

Section I – 2 marks

- Allow about 3 minutes for this section

Section II – 39 marks

- Allow about 57 minutes for this section

Section I (2 marks)	Multiple Choice	/2
Section II (39 marks)	Question 3	/10
	Question 4	/11
	Question 5	/9
	Question 6	/9
Total		/41

This question paper must not be removed from the examination room.

This assessment task constitutes 25% of the course.

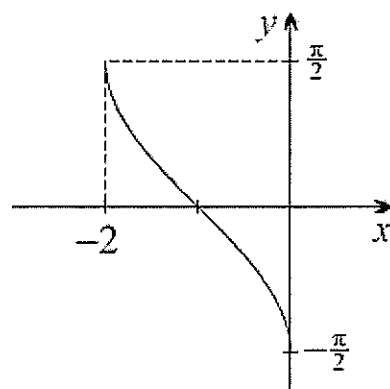
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Question 1

The exact value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is:

- (A) $\frac{5\pi}{6}$
- (B) $-\frac{5\pi}{6}$
- (C) $-\frac{\pi}{6}$
- (D) $\frac{\pi}{3}$

Question 2



The diagram above shows the graph of

- (A) $y = \sin^{-1}(x + 1)$
- (B) $y = \sin^{-1}(x - 1)$
- (C) $y = \cos^{-1}(x + 1) - \frac{\pi}{2}$
- (D) $y = \cos^{-1}(x - 1) - \frac{\pi}{2}$

Question 3 (10 marks) Use a **SEPARATE** writing booklet

- (a) Find the derivative of $y = \cos^{-1}(2x)$ 1
- (b) Evaluate $\int_0^1 \frac{1}{\sqrt{4-x^2}} dx$. 2
- (c) State the domain and range of $y = 3 \cos^{-1} \frac{x}{2}$? 2
- (d) Find the exact value of $\cos \left[\sin^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{4}{3} \right) \right]$ 2
- (e)
- i) Show that $\frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2}) = \cos^{-1} x$ 2
- ii) Hence evaluate $\int_0^1 \cos^{-1} x dx$ 1

Question 4 (11 marks) Use a **SEPARATE** writing booklet

- (a) How many arrangements of the word “excellent” are possible:
- i) if there are no restrictions? 2
 - ii) the word starts and end with an “l” ? 1
 - iii) the three “e” ’s are together ? 2
- (b) A committee of five, is to be chosen from 7 New South Welshman and 6 Queenslanders:
- i) How many committee’s can be chosen? 1
 - ii) What is the probability that the committee has 5 Queenslanders? 1
- (c) 8 men and 8 women are to be seated at a round table.
- i) In how many ways can they be seated if men and women alternate? 2
 - ii) A particular man and a woman refuse to sit next to each other. How many ways can the group now be seated at the round table, if men and women alternate? 2

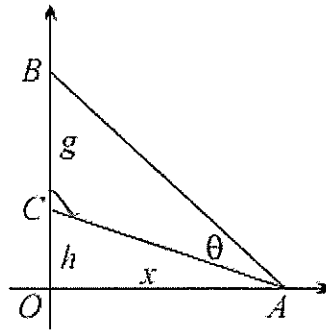
Question 5 (9 marks) Use a **SEPARATE** writing booklet

- (a) Prove ${}^nC_k = {}^{n-1}C_{k-1} + {}^{n-1}C_k$ **3**
(Note: this proof assumes that $1 \leq k \leq n-1$)
- (b) Prove by Induction that the sum of the first n powers of 3 (starting with $n = 0$) is given **3**
by:
$$3^0 + 3^1 + 3^2 + \dots + 3^n = \frac{1}{2}(3^{n+1} - 1)$$
- (c) Prove by Induction that $3^{2n} - 1$ is divisible by 8, for all $n \geq 1$ **3**

Question 6 (9 marks) Use a SEPARATE writing booklet

- (a) Let $f(x) = 3 + 2x - x^2$. For $x \leq 1$ this function has an inverse $f^{-1}(x)$.
- i) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same number plane. 2
- ii) Find an expression for $f^{-1}(x)$. 2

(b)

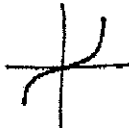
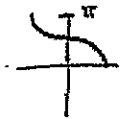


A billboard of height g metres is mounted on a wall at BC which is h metres above level ground at O. The billboard subtends an angle θ when observed at ground level at A, x metres from O. $\angle BOA$ is a right angle.

- i) Show that $\theta = \tan^{-1}\left(\frac{g+h}{x}\right) - \tan^{-1}\left(\frac{h}{x}\right)$. 1
- ii) Show that θ is a maximum when $x = \sqrt{h(g+h)}$. 2
- iii) It is known that the maximum value of θ is $\frac{\pi}{4}$, show that $g = 2h(1 + \sqrt{2})$. 2

End of Examination

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1a $\frac{3}{4}$ $y = \tan^{-1} x$ for $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ $\therefore \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ $= -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ $= -\frac{\pi}{6} \therefore \underline{C}$	
2 $\sin^{-1} x$  $\cos^{-1} x$  $\therefore \cos^{-1} x$ variant centre of graph shifted to $x = -1$ $\therefore \cos^{-1}(x+1) = \frac{\pi}{2}$ $\therefore \underline{C}$	

Question 3

(a) $\frac{dy}{dx} = -\frac{2}{\sqrt{1-4x^2}}$ MUST USE REFERENCE SHEET	1-correct answer	
b) $\int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_0^1 = \frac{\pi}{6}$	1-correct integration 1-correct answer	

c) Domain: $-1 \leq \frac{x}{2} \leq 1$
 $-2 \leq x \leq 2$

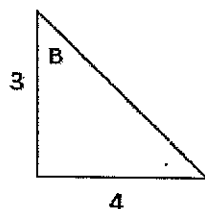
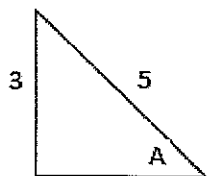
Range: $0 \leq y \leq 3\pi$

2-correct domain & range

d) $\cos \left[\sin^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{4}{3} \right) \right]$

Let $A = \sin^{-1} \left(\frac{3}{5} \right)$

$B = \tan^{-1} \left(\frac{4}{3} \right)$



$$\begin{aligned} \cos(A - B) &= \cos A \cdot \cos B + \sin A \cdot \sin B \\ &= \frac{4}{5} \cdot \frac{3}{5} + \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25} \end{aligned}$$

1-correct expressions of triangles

1-correct answer with working

e)

(i) $\frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2}) = 1 \cdot \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} - \frac{1}{2} \times -\frac{2x}{\sqrt{1-x^2}}$
 $= \cos^{-1} x$

2-correct derivative of both functions with working

(ii) $\int_0^1 \cos^{-1} x \, dx = [x \cos^{-1} x - \sqrt{1-x^2}]_0^1 = 1$

1-correct answer

$$-\frac{2}{\sqrt{1-4x^2}}$$

Question 4

a

$$i) = \frac{9!}{3!2!} = 30,240$$

ii) $\boxed{2} \square \square \square \square \square \square \square \boxed{2}$

$$= \frac{7!}{3!} = 840 \quad (\text{have to account for 3 x's})$$

iii) $\boxed{3 \times 5} \square \square \square \square \square \square$

$$= \frac{7!}{2!} \quad (\text{have to account for 2 x's})$$

$$= 2520$$

b 7 NSW, 6 QLD = 13 intotal, committee of 5

$$i) {}^{13}C_5 = 1287$$

ii) to pick 5 Q'landers from 6 = 6C_5

$$\therefore P(5 \text{ Q'landers}) = \frac{{}^6C_5}{{}^{13}C_5} = \frac{6}{1287} = \frac{2}{429}$$

2 marks: Correct answer.

1 mark: Any one of the repetition is missing

Correct answer with correct working
1 mark.

2 marks: Correct answer from correct working

1 mark: $\frac{7!}{3!2!}$ (minor error)

1 mark

1 mark
(Working required)

well done.

Well done.

Generally well done

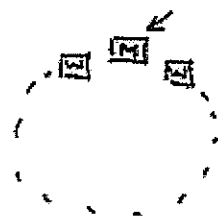
Some students did this error.

Well done.

Well done.

4c

8m, 8w, round table



imagine locking in 1 man
7m left in $7!$ ways
8w left in $8!$ ways

$\therefore 7! \times 8!$ as if we lock a woman
in 1st, we get the same
arrangement

$$\therefore 7! \times 8! = 203212800$$

2 marks:

Correct answer
from correct
working

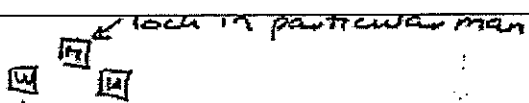
$7! \times 7!$ received
1 mark.

NOTE: explanation is important
to get part marks.

Many students fixed
one person and got
 $1 \times 7!$ correct.

However, they forget
the 8 women can
take any of the
8 positions, thus
 $8!$

/c:



7 men left in $7!$ ways
6 women left in $6!$ ways

$$\therefore 2 \times {}^7C_2 \times 7! \times 6! = 76,204,800 \times 2 \\ = 152,409,600$$

OR

$$\text{lock (H, W) together} \\ = 2 \times 7! \times 7! = 50,803,200$$

$$\therefore \text{Not together} \\ = 203,212,800 - 50,803,200 \\ = 152,409,600$$

2 Marks :

Correct answer
from correct working.

$$\text{Ans (i)} - 2 \times 7! \times 7!$$

1 mark : Subtracting

$$\text{Incorrect } 2 \times 7! \times 7!$$

from answer (i)

Students had extreme
difficulty with this
question

Question 5

using ${}^nC_r = \frac{n!}{(n-r)!r!}$

$\therefore \text{LHS} = \frac{n!}{(n-k)!k!}$

$\text{RHS} = \frac{(n-1)!}{[(n-1)-(k-1)]!(k-1)!} + \frac{(n-1)!}{(n-1-k)!k!}$

$= \frac{(n-1)!}{(n-k)!(k-1)!} + \frac{(n-1)!}{(n-k-1)!k!}$

[1 mark]

need common denom'r of $(n-k)!k!$

$= \frac{(n-1)!}{(n-k)!(k-1)!} \times \frac{k}{k} + \frac{(n-1)!}{(n-k-1)!k!} \times \frac{(n-k)}{(n-k)}$

[1 mark]

$= \frac{k(n-1)! + n \times (n-1)! - k(n-1)!}{(n-k)!k!}$

$= \frac{n!}{(n-k)!k!} = \text{LHS}$

} [1 mark]

5b

Prove $3^0 + 3^1 + 3^2 + \dots + 3^n = \frac{1}{2}(3^{n+1} - 1)$

when $n=0$

$$3^0 = 1 = \text{LHS}$$

$$\text{RHS} = \frac{1}{2}(3^{0+1} - 1) = \frac{1}{2}(3 - 1) = \frac{1}{2} \cdot 2 = 1 = \text{LHS} \quad \checkmark$$

assume true for $n=k$

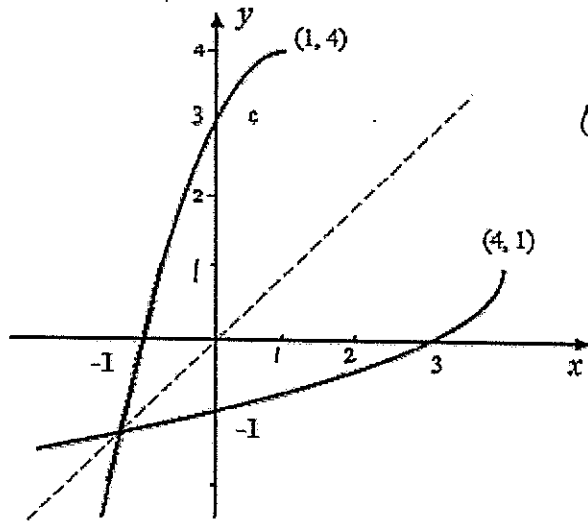
$$3^0 + 3^1 + 3^2 + \dots + 3^k = \frac{1}{2}(3^{k+1} - 1)$$

prove true for $n=k+1$

$$\underbrace{3^0 + 3^1 + 3^2 + \dots + 3^k}_{\text{using expression for } n=k} + 3^{k+1} = \frac{1}{2}[3^{(k+1)+1} - 1]$$

Question 6

(a) (i)



$$\begin{aligned} (ii) \quad x &= 3 + 2y - y^2 \\ y^2 - 2y - (3 - x) &= 0 \\ y &= \frac{2 \pm \sqrt{4 + 4(3 - x)}}{2} \\ &= \frac{2 \pm \sqrt{16 - 4x}}{2} = 1 \pm \sqrt{4 - x} \\ \therefore y &= 1 - \sqrt{4 - x} \quad y \leq 1 \end{aligned}$$

$$\begin{aligned} \text{or} \\ y^2 - 2y &= 3 - x \\ y^2 - 2y + 1 &= 4 - x \\ (y - 1)^2 &= 4 - x \\ y &= 1 \pm \sqrt{4 - x} \end{aligned}$$

(b)

$$\theta = \angle OAB - \angle OAC$$

$$\tan \angle OAB = \frac{g+h}{x} \rightarrow \angle OAB = \tan^{-1} \left(\frac{g+h}{x} \right)$$

$$\tan \angle OAC = \frac{h}{x} \rightarrow \angle OCA = \tan^{-1} \left(\frac{h}{x} \right)$$

$$\therefore \theta = \tan^{-1} \left(\frac{g+h}{x} \right) - \tan^{-1} \left(\frac{h}{x} \right)$$

(1) 1 mark.

$$\frac{d\theta}{dx} = \frac{1}{1 + \frac{(g+h)^2}{x^2}} \times -\frac{g+h}{x^2} + \frac{1}{1 + \frac{h^2}{x^2}} \times -\frac{h}{x^2} = -\frac{g+h}{x^2 + (g+h)^2} + \frac{h}{x^2 + h^2}$$

$$\text{For max. } \theta, \quad \frac{d\theta}{dx} = 0 \rightarrow -\frac{g+h}{x^2 + (g+h)^2} + \frac{h}{x^2 + h^2} = 0$$

$$\begin{aligned} -(g+h)(x^2 + h^2) + h(x^2 + (g+h)^2) &= 0 \\ -gx^2 - gh^2 - hx^2 - h^3 + hx^2 + hg^2 + 2gh^2 + h^3 &= 0 \\ -gx^2 + hg^2 + gh^2 &= 0 \end{aligned}$$

$$\therefore x^2 = \frac{hg(g+h)}{g} \rightarrow x = \sqrt{h(g+h)} \quad (x > 0) \quad 1 \text{ mark.}$$

$$\frac{d^2\theta}{dx^2} = -\frac{2x(g+h)}{(x^2 + (g+h)^2)^2} - \frac{2xh}{(x^2 + h^2)^2} < 0 \quad \therefore \theta \text{ is a maximum.} \quad 1 \text{ mark.}$$

1-correct shape

1-correct labelling

2- correct answer with working

1 mark: $1 \pm \sqrt{4-x}$.

1-correct answer with full working shown

1-correct derivative

1-correct answer with working

(1) When sketching graphs, you must show scales and labels on axes. Label major features. (x, y) changed to (y, x) on the inverse must be clearly shown. Put the $y=x$ line in.

Students continue to have difficulty using quadratic formula & completing the square

• Most students received 1 mark for finding the expression for θ .

• Some students experienced difficulty differentiating $\frac{g+h}{x}$, $\frac{h}{x}$.

• Very few students proved the x -value was a maximum. $\frac{d\theta}{dx} = 0$ gives you stationary point

Sub. $x = \sqrt{h(g+h)}$ and $\theta = \frac{\pi}{4}$ into (1)

$$\frac{\pi}{4} = \tan^{-1}\left(\frac{g+h}{\sqrt{h(g+h)}}\right) - \tan^{-1}\left(\frac{h}{\sqrt{h(g+h)}}\right)$$

$$\rightarrow 1 = \frac{\frac{g+h}{\sqrt{h(g+h)}} - \frac{h}{\sqrt{h(g+h)}}}{1 + \frac{h(g+h)}{h(g+h)}} \rightarrow \sqrt{\frac{g+h}{h}} - \sqrt{\frac{h}{g+h}} = 2 \text{ square both sides}$$

$$\rightarrow \frac{g+h}{h} - 2 + \frac{h}{g+h} = 4 \rightarrow \frac{g+h}{h} - \frac{h}{g+h} = 6$$

$$(g+h)^2 + h^2 = 6h(g+h) \quad \text{completing the square}$$

$$(g+h)^2 - 6h(g+h) + 9h^2 = 8h^2$$

$$((g+h) - 3h)^2 = 8h^2 \rightarrow (g-2h)^2 = 8h^2$$

$$g-2h = 2\sqrt{2}h \rightarrow g = 2h(1 + \sqrt{2})$$

1-showing the substitution of x & θ .

1-correct answer with working

Most students received the first mark.

5b

Prove $3^0 + 3^1 + 3^2 + \dots + 3^n = \frac{1}{2}(3^{n+1} - 1)$

when $n=0$

$$3^0 = 1 = \text{LHS}$$

$$\text{RHS} = \frac{1}{2}(3^{0+1} - 1) = \frac{1}{2}(3 - 1) = \frac{1}{2} \cdot 2 = 1 = \text{LHS} \checkmark$$

} [1 mark]

assume true for $n=k$

$$3^0 + 3^1 + 3^2 + \dots + 3^k = \frac{1}{2}(3^{k+1} - 1)$$

prove true for $n=k+1$

$$\underbrace{3^0 + 3^1 + 3^2 + \dots + 3^k}_{\text{using expression for } n=k} + 3^{k+1} = \frac{1}{2}[3^{(k+1)+1} - 1]$$

using expression for $n=k$

$$\frac{1}{2}(3^{k+1} - 1) + 3^{k+1} = \frac{1}{2}(3^{k+2} - 1) \quad [1 \text{ mark}]$$

with LHS

$$= \frac{1}{2}(3^{k+1} - 1) + \frac{2 \cdot 3^{k+1}}{2}$$

$$= \frac{3 \cdot 3^{k+1} - 1}{2}$$

$$= \frac{1}{2}(3^{k+2} - 1) = \text{RHS}$$

} [1 mark]

$\therefore$ as true for $n=0$, assumed true for $n=k$,
and proven true for $n=k+1$

$\therefore$ By induction it must be true for $n=0, 1, 2, \dots$

$\therefore$ true for all $n \geq 0$.

5c Prove $3^{2n} - 1$ divisible by 8 for $n \geq 1$

when $n=1$

$$3^{2 \times 1} - 1 = 3^2 - 1 = 9 - 1 = 8$$

$$8 \div 8 = 1 \quad \therefore \text{true for } n=1$$

[1 mark]

assume true for $n=k$

$$3^{2k} - 1 = 8a \quad (\text{where } a \text{ is an integer})$$

> 0

prove true for $n=k+1$

$$3^{2(k+1)} - 1 = 8b \quad (\text{where } b \text{ is a positive integer})$$

$$\text{LHS} = 3^{2k+2} - 1$$

$$= 3^{2k} \cdot 3^2 - 1$$

from $n=k$ assumption

$$3^{2k} = 8a + 1$$

sub into LHS

$$= (8a+1)9 - 1$$

[1 mark]

$$= 72a + 9 - 1$$

$$= 72a + 8$$

$$= 8(1+9a) = 8b$$

$\therefore$ true for $n=k+1$

[1 mark]

$\therefore$ as true for $n=1$, assumed true for $n=k$,

and proven true for $n=k+1$,

by induction it is true for $n=2, 3$, etc... $\therefore$ true for all $n \geq 1$.

Q5 Exam Feedback

a many students had forgotten ${}^nC_k = \frac{n!}{(n-k)!k!}$

- some tried to prove by induction & got in a mess.
- many didn't get a common denominator, or got a HUGE common denominator that made the question more difficult.
- some skipped lines of working & lost a mark.
- most students need to revise.

b was quite well done overall.

- some tested $n=1$, not $n=0$
- most did $n=0$, and substituted $n=k$ expression correctly, some need to work on Algebra skills / index skills as couldn't complete the question

c mostly very well done.

- some forgot to have a different pronumeral for 8 times e.g. $8a$, $8b$ or $8Q$, $8M$ at $n=k$, $n=k+1$ stages.
- most got full marks
- some silly arithmetic / manipulation errors.

Note: for part b, c $\Rightarrow$ some students didn't conclude the proof correctly and a mark was deducted.